

Geology Lessons





About the Course

An elective in Earth Science, Geology is a wonderful course for Alveary high school students who are interested in the history, theory, and methods of the field.

This topic is included in the following course(s): Science: Geology



Placement & Combining Tips

Due to the complexity of certain topics and the maturity required, this course is recommended for Grades 11-12.



Scheduling

GRADE	SCHEDULE INFO.	BOOKS
11-12	Geology Lessons 5 times/week 45 min	101 American Geo-Sites You've Gotta See Annals of the Former World Roadside Geology/Geology Underfoot Series The Field Guide to Geology The Map That Changed the World The Story of the Earth in 25 Rocks

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Geology				
Geology Lessons Nature Walks & Scouting: Grades 9-12	Geology Lessons	Geology Lessons	Geology Lessons Nature Notebook: Grades 9-12	Geology Lessons Geology Labs



Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Geology Lessons

- ☐ Carve out time to continue or establish the regular habit of spending time in nature, including the use of a nature journal, as appropriate.
- ☐ Obtain materials from the supply lists.
- ☐ Select a science book from the Alveary bookshelf for personal reading time, as appropriate.

☐ Bookmark or print Quick Links, as needed.



Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

Geology Lessons



101 American Geo-Sites You've Gotta See



Annals of the Former World



Roadside Geology/Geology Underfoot Series



The Field Guide to Geology



The Map That Changed the World



The Story of the Earth in 25 Rocks



Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

(No Subject Supplies Assigned)



Quick Links

Geology Lessons

∞ [Nature Journal Connection](#)

∞ [Alveary Bookshelf](#)

∞ [Roadside Geology](#)

∞ [State by State Rockhound](#)

∞ [How to Find Rocks](#)

[Click THIS text
or scan the QR
code for links.](#)



Geology Lessons

How To Approach



Introduce

- If starting a new book or a new topic in the book, then look at the title or a picture and take a moment to consider previous ideas and experiences.
- If continuing a previous reading, recap what was read previously. Often, the title of the book's section can help to draw out the main idea.



Read

- Read or do, as instructed in the lessons, making note of or flagging unfamiliar terms, interesting ideas, important dates, inspiring quotes, etc.
- Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.
- As they read, learners record ideas in a notebook or binder using outlines, diagrams, graphic organizers, or other methods (or a combination of methods) that suit them. These recordings can be a helpful mechanism for remembering or a mini-narration to support understanding.
- If learners do not understand a word or concept, do not worry. Try reading over the passage again, studying a picture or diagram, connecting the idea to something from real life, or practicing chapter exercises. The lab/field work further supports major concepts from the text.



Narrate

- Process the ideas of the lesson by retelling, defining a concept, explaining the links in a chain of thought, etc. Do this orally or silently to yourself.
- Use words, pictures, outlines, etc.
- If a particular idea cannot be narrated, then read or examine the text again.



Discuss

- Consider with the teacher any thoughts, confusion, or concerns about the passage.
- If understanding is still uncertain, try rereading the passage or do some personal research on the topic.



Connect

- Follow any extra links, examine any sidebars in the text, or pursue additional reading, depending on student interest.



Term 1

WEEK 1 45m Geology Lessons - Lesson 1

How Geology Works

Materials: The Field Guide to Geology

→ NOTE

If you have a rock and/or fossil kit, you may find it useful to keep it nearby throughout this course. As different materials are mentioned in your various books, you can reference these materials in your kit(s).

→ READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

• PLAN WEEKLY

- nature walk - record observations
- science free read

WEEK 1 45m Geology Lessons - Lesson 2

Practical Interpretation

Materials: Roadside Geology & Notebook

→ READ, NARRATE, & DISCUSS

Divide up your state or regional geology book so that you finish it this term. (If you are not able to finish it in one term, that is okay. You might choose to continue it next term or to leave off.) As you read, map your state or region, showing the interesting geological features. Plan field trips, if possible, to some of the sites. Record your observations and findings.

WEEK 1 45m Geology Lessons - Lesson 3

Identification

→ INTRO

This is your day each week to prepare for your field studies using a mixture of resources provided for help, time for planning, and to spend time with the rocks you have found.

→ RESEARCH

Start at Gator Girl Rocks to begin your journey. The How to Find Rocks website is helpful, as well. You will want to read about the laws that pertain to rockhounding and visit the links under your state. Read about any local resources, and begin making a plan. You might want to visit local nature centers and museums, and ask about local geology guides or guided tours. You might want to track down any local clubs or groups to join. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

∞ Link: Gator Girl Rocks

∞ Link: How to Find Rocks

∞ Link: Rock Clubs by State

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.



Term 1

WEEK 1 45m Geology Lessons - Lesson 4

Geology Literature

Materials: Annals of the Former World

→ INTRO

People say this book is the "Literature of Geology". See what you think by the end of the term. McPhee travels across the country with various geologists, writing about the geological wonders along the way. He is writing for people who are not geologists, but that doesn't mean that the text is easy. Take in what you can and enjoy the views out the window.

This is a big book. Read Book 1 (Basin and Range), and then decide if you want to keep reading in order, read about places near you, or read about ones that you have never visited. Go at your own pace. If you read and narrate around 20 pages per week, you will get a good way through the book by the end of Term 2. Then you can decide if you want to finish as a free read or not.

→ READ, NARRATE, & DISCUSS

Be sure to look at the map on p.2-3 and read the Narrative Table of Contents before plunging into Chapter 1. Refer to the maps and charts in the text and at the end of the book while you are reading. You may want to look up unfamiliar terms as you read and make a collection of geological terms. Come up with an interesting way to narrate, perhaps an illustrated map or travel journal?

~ 20 pages

WEEK 1 45m Geology Lessons - Lesson 5

Flex Day/Current Events

Materials: Selected geology literature, lab/field supplies, or internet access

→ INTRO

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (linked in the lesson and in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as The Map that Changed the World, or more active by pursuing a special study of interest.

∞ Article Link: Geology Current Events

WEEK 2 45m Geology Lessons - Lesson 6

How Geology Works

Materials: The Field Guide to Geology

→ READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 2 45m Geology Lessons - Lesson 7

Practical Interpretation

Materials: Roadside Geology & Notebook

→ READ, NARRATE, & DISCUSS

Continue reading. As you read, map your state or region, showing the interesting geological features. Plan field trips, if possible, to some of the sites. Record your observations and findings.



Term 1

WEEK 2 ☐ 45m Geology Lessons - Lesson 8

Identification

→ RESEARCH

Google your state's Geological Survey to see if they have any online resources that would be helpful to you, or check out the USGS maps online. If an office is nearby, they may have some maps or other resources you could pick up. Also, see if they will answer any of your questions about your local geology. If not, see if they know someone you could talk to.

∞ Website Link: USGS Online Resources

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 2 ☐ 45m Geology Lessons - Lesson 9

Geology Literature

☐ Materials: Annals of the Former World

→ READ, NARRATE, & DISCUSS

~ 20 pages

WEEK 2 ☐ 45m Geology Lessons - Lesson 10

Flex Day/Current Events

☐ Materials: Selected geology literature, lab/field supplies, or internet access

→ NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as The Map that Changed the World, or more active by pursuing a special study of interest.

WEEK 3 ☐ 45m Geology Lessons - Lesson 11

How Geology Works

☐ Materials: The Field Guide to Geology

→ READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 3 ☐ 45m Geology Lessons - Lesson 12

Practical Interpretation

☐ Materials: Roadside Geology & Notebook

→ READ, NARRATE, & DISCUSS

Continue reading. As you read, map your state or region, showing the interesting geological features. Plan field trips, if possible, to some of the sites. Record your observations and findings.



Term 1

WEEK 3 ☐ 45m Geology Lessons - Lesson 13

Identification

→ RESEARCH

Spend some more time considering the various geological maps. They can take some getting used to, so be patient!

∞ Website Link: USGS Online Resources

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 3 ☐ 45m Geology Lessons - Lesson 14

Geology Literature

☐ Materials: Annals of the Former World

→ READ, NARRATE, & DISCUSS

~ 20 pages

WEEK 3 ☐ 45m Geology Lessons - Lesson 15

Flex Day/Current Events

☐ Materials: Selected geology literature, lab/field supplies, or internet access

→ NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

WEEK 4 ☐ 45m Geology Lessons - Lesson 16

How Geology Works

☐ Materials: The Field Guide to Geology

→ READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 4 ☐ 45m Geology Lessons - Lesson 17

Practical Interpretation

☐ Materials: Roadside Geology

→ READ, NARRATE, & DISCUSS

Continue reading. As you read, map your state or region, showing the interesting geological features. Plan field trips, if possible, to some of the sites. Record your observations and findings.



Term 1

WEEK 4 45m Geology Lessons - Lesson 18

Identification

Materials: rock/mineral/fossil guides, painting tools of your choice & Nature Notebook

→ RESEARCH

Start trying to identify any rocks, minerals, or fossils that you have picked up locally. Be sure to paint some of your rock finds in your Nature Notebook as you go through the term.

∞ Video Link: How to Draw Rocks

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 4 45m Geology Lessons - Lesson 19

Geology Literature

Materials: Annals of the Former World

→ READ, NARRATE, & DISCUSS

~ 20 pages

WEEK 4 45m Geology Lessons - Lesson 20

Flex Day/Current Events

Materials: Selected geology literature, lab/field supplies, or internet access

→ NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as The Map that Changed the World, or more active by pursuing a special study of interest.

WEEK 5 45m Geology Lessons - Lesson 21

How Geology Works

Materials: The Field Guide to Geology

→ READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 5 45m Geology Lessons - Lesson 22

Practical Interpretation

Materials: Roadside Geology & Notebook

→ READ, NARRATE, & DISCUSS

Continue reading. As you read, map your state or region, showing the interesting geological features. Plan field trips, if possible, to some of the sites. Record your observations and findings.



Term 1

WEEK 5 45m Geology Lessons - Lesson 23

Identification

Materials: rock/mineral/fossil guides

→ RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day. Try to plan at least 3 field trips over the course of the year: use your books or local resources for inspiration.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 5 45m Geology Lessons - Lesson 24

Geology Literature

Materials: Annals of the Former World

→ READ, NARRATE, & DISCUSS

~ 20 pages

WEEK 5 45m Geology Lessons - Lesson 25

Flex Day/Current Events

Materials: Selected geology literature, lab/field supplies, or internet access

→ NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as The Map that Changed the World, or more active by pursuing a special study of interest.

WEEK 6 45m Geology Lessons - Lesson 26

How Geology Works

Materials: The Field Guide to Geology

→ READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 6 45m Geology Lessons - Lesson 27

Practical Interpretation

Materials: Roadside Geology

→ READ, NARRATE, & DISCUSS

Continue reading. As you read, map your state or region, showing the interesting geological features. Plan field trips, if possible, to some of the sites. Record your observations and findings.



Term 1

WEEK 6 45m Geology Lessons - Lesson 28

Identification

Materials: rock/mineral/fossil guides

→ RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day. Try to plan at least 3 field trips over the course of the year: use your books or local resources for inspiration.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 6 45m Geology Lessons - Lesson 29

Geology Literature

Materials: Annals of the Former World

→ READ, NARRATE, & DISCUSS

~ 20 pages

WEEK 6 45m Geology Lessons - Lesson 30

Flex Day/Current Events

Materials: Selected geology literature, lab/field supplies, or internet access

→ NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as The Map that Changed the World, or more active by pursuing a special study of interest.

WEEK 7 45m Geology Lessons - Lesson 31

How Geology Works

Materials: The Field Guide to Geology

→ READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 7 45m Geology Lessons - Lesson 32

Practical Interpretation

Materials: Roadside Geology & Notebook

→ READ, NARRATE, & DISCUSS

Continue reading. As you read, map your state or region, showing the interesting geological features. Plan field trips, if possible, to some of the sites. Record your observations and findings.



Term 1

WEEK 7 45m Geology Lessons - Lesson 33

Identification

Materials: rock/mineral/fossil guides

→ RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 7 45m Geology Lessons - Lesson 34

Geology Literature

Materials: Annals of the Former World

→ READ, NARRATE, & DISCUSS

~ 20 pages

WEEK 7 45m Geology Lessons - Lesson 35

Flex Day/Current Events

Materials: Selected geology literature, lab/field supplies, or internet access

→ NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as The Map that Changed the World, or more active by pursuing a special study of interest.

WEEK 8 45m Geology Lessons - Lesson 36

How Geology Works

Materials: The Field Guide to Geology

→ READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 8 45m Geology Lessons - Lesson 37

Practical Interpretation

Materials: Roadside Geology & Notebook

→ READ, NARRATE, & DISCUSS

Continue reading. As you read, map your state or region, showing the interesting geological features. Plan field trips, if possible, to some of the sites. Record your observations and findings.



Term 1

WEEK 8 45m Geology Lessons - Lesson 38

Identification

Materials: rock/mineral/fossil guides

→ RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 8 45m Geology Lessons - Lesson 39

Geology Literature

Materials: Annals of the Former World

→ READ, NARRATE, & DISCUSS

~ 20 pages

WEEK 8 45m Geology Lessons - Lesson 40

Flex Day/Current Events

Materials: Selected geology literature, lab/field supplies, or internet access

→ NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as The Map that Changed the World, or more active by pursuing a special study of interest.

WEEK 9 45m Geology Lessons - Lesson 41

How Geology Works

Materials: The Field Guide to Geology

→ READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 9 45m Geology Lessons - Lesson 42

Practical Interpretation

Materials: Roadside Geology & notebook

→ READ, NARRATE, & DISCUSS

Continue reading. As you read, map your state or region, showing the interesting geological features. Plan field trips, if possible, to some of the sites. Record your observations and findings.



Term 1

WEEK 9 45m Geology Lessons - Lesson 43

Identification

Materials: rock/mineral/fossil guides

→ RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 9 45m Geology Lessons - Lesson 44

Geology Literature

Materials: Annals of the Former World

→ READ, NARRATE, & DISCUSS

~ 20 pages

WEEK 9 45m Geology Lessons - Lesson 45

Flex Day/Current Events

Materials: Selected geology literature, lab/field supplies, or internet access

→ NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as The Map that Changed the World, or more active by pursuing a special study of interest.

WEEK 10 45m Geology Lessons - Lesson 46

How Geology Works

Materials: The Field Guide to Geology

→ READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 10 45m Geology Lessons - Lesson 47

Practical Interpretation

Materials: Roadside Geology & Notebook

→ READ, NARRATE, & DISCUSS

Continue reading. As you read, map your state or region, showing the interesting geological features. Plan field trips, if possible, to some of the sites. Record your observations and findings.



Term 1

WEEK 10 45m Geology Lessons - Lesson 48

Identification

Materials: rock/mineral/fossil guides

→ RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 10 45m Geology Lessons - Lesson 49

Geology Literature

Materials: Annals of the Former World

→ READ, NARRATE, & DISCUSS

~ 20 pages

WEEK 10 45m Geology Lessons - Lesson 50

Flex Day/Current Events

Materials: Selected geology literature, lab/field supplies, or internet access

→ NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as The Map that Changed the World, or more active by pursuing a special study of interest.

WEEK 11 45m Geology Lessons - Lesson 51

How Geology Works

Materials: The Field Guide to Geology

→ READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 11 45m Geology Lessons - Lesson 52

Practical Interpretation

Materials: Roadside Geology & Notebook

→ READ, NARRATE, & DISCUSS

Continue reading. As you read, map your state or region, showing the interesting geological features. Plan field trips, if possible, to some of the sites. Record your observations and findings.



Term 1

WEEK 11 45m Geology Lessons - Lesson 53

Identification

Materials: rock/mineral/fossil guides

→ RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally.

WEEK 11 45m Geology Lessons - Lesson 54

Geology Literature

Materials: Annals of the Former World

→ READ, NARRATE, & DISCUSS

~ 20 pages

WEEK 11 45m Geology Lessons - Lesson 55

Present What You've Learned

→ PREPARE

For one of your exam questions next week, you will be asked to give an informal presentation to your teacher on what you have discovered so far about your local geology and geological history. Use this time to gather your thoughts and to think through how you want to do that. You can be as creative as you would like. You will do a more formal presentation at the end of the year. This is more like an update on what you have discovered so far.

WEEK 12 45m Geology Lessons - Lesson 56

Exam Week

→ EXAMS

Answer question(s) related to course.

WEEK 12 45m Geology Lessons - Lesson 57

Exam Week

→ EXAMS

Answer question(s) related to course.

WEEK 12 45m Geology Lessons - Lesson 58

Exam Week

→ EXAMS

Answer question(s) related to course.

WEEK 12 45m Geology Lessons - Lesson 59

Exam Week

→ EXAMS

Answer question(s) related to course.

WEEK 12 45m Geology Lessons - Lesson 60

Exam Week

→ EXAMS

Answer question(s) related to course.



Term 2

WEEK 1

45m Geology Lessons - Lesson 61

How Geology Works

Materials: The Field Guide to Geology

→ READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 1

45m Geology Lessons - Lesson 62

Practical Interpretation

Materials: 101 American Geo-Sites You've Gotta See & Notebook

→ INTRO

You read about your state or region last term; this term, we will expand the survey to the country's borders.

→ READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

WEEK 1

45m Geology Lessons - Lesson 63

Identification

→ RESEARCH

Where do you plan to visit this term? Is there a specific rock type or formation that you want to learn more about? Use the resources that you discovered last term to make a plan for this term. What questions about your local geology do you have? How will you find the answers?

- ∞ Link: Gator Girl Rocks
- ∞ Link: How to Find Rocks
- ∞ Link: Rock Clubs by State

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day. Try to plan at least 3 field trips over the course of the year: use your books or local resources for inspiration.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 1

45m Geology Lessons - Lesson 64

Geology Literature

Materials: Annals of the Former World

→ READ, NARRATE, & DISCUSS

~ 20 pages

WEEK 1

45m Geology Lessons - Lesson 65

Flex Day/Current Events

Materials: Selected geology literature, lab/field supplies, or internet access

→ NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some



Term 2

geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as The Map that Changed the World, or more active by pursuing a special study of interest.

WEEK 2

45m Geology Lessons - Lesson 66

How Geology Works

Materials: The Field Guide to Geology

→ READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 2

45m Geology Lessons - Lesson 67

Practical Interpretation

Materials: 101 American Geo-Sites You've Gotta See & Notebook

→ READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

WEEK 2

45m Geology Lessons - Lesson 68

Identification

Materials: rock/mineral/fossil guides

→ RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day. Try to plan at least 3 field trips over the course of the year: use your books or local resources for inspiration.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 2

45m Geology Lessons - Lesson 69

Geology Literature

Materials: Annals of the Former World

→ READ, NARRATE, & DISCUSS

~ 20 pages

WEEK 2

45m Geology Lessons - Lesson 70

Flex Day/Current Events

Materials: Selected geology literature, lab/field supplies, or internet access

→ NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some



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geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

WEEK 3

45m Geology Lessons - Lesson 71

How Geology Works

Materials: The Field Guide to Geology

→ READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 3

45m Geology Lessons - Lesson 72

Practical Interpretation

Materials: 101 American Geo-Sites You've Gotta See & Notebook

→ READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

WEEK 3

45m Geology Lessons - Lesson 73

Identification

Materials: rock/mineral/fossil guides

→ RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Consider taking part in a citizen science project on your field days:

- ∞ Link: Citizen Science Sand Project
- ∞ Link: Chronolog

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 3

45m Geology Lessons - Lesson 74

Geology Literature

Materials: Annals of the Former World

→ READ, NARRATE, & DISCUSS

~ 20 pages

WEEK 3

45m Geology Lessons - Lesson 75

Flex Day/Current Events

Materials: Selected geology literature, lab/field supplies, or internet access

→ NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some



Term 2

geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

WEEK 4

45m Geology Lessons - Lesson 76

How Geology Works

Materials: The Field Guide to Geology

→ READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 4

45m Geology Lessons - Lesson 77

Practical Interpretation

Materials: 101 American Geo-Sites You've Gotta See & Notebook

→ READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

WEEK 4

45m Geology Lessons - Lesson 78

Identification

Materials: rock/mineral/fossil guides

→ RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. If you haven't looked at them under magnification yet, give it a try! Read the linked article for inspiration:

∞ Article Link: Sand Grains

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 4

45m Geology Lessons - Lesson 79

Geology Literature

Materials: Annals of the Former World

→ READ, NARRATE, & DISCUSS

~ 20 pages

WEEK 4

45m Geology Lessons - Lesson 80

Flex Day/Current Events

Materials: Selected geology literature, lab/field supplies, or internet access

→ NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some



Term 2

geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as The Map that Changed the World, or more active by pursuing a special study of interest.

WEEK 5

45m Geology Lessons - Lesson 81

How Geology Works

Materials: The Field Guide to Geology

→ READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 5

45m Geology Lessons - Lesson 82

Practical Interpretation

Materials: 101 American Geo-Sites You've Gotta See & Notebook

→ READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

WEEK 5

45m Geology Lessons - Lesson 83

Identification

Materials: rock/mineral/fossil guides

→ RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day. Try to plan at least 3 field trips over the course of the year: use your books or local resources for inspiration.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 5

45m Geology Lessons - Lesson 84

Geology Literature

Materials: Annals of the Former World

→ READ, NARRATE, & DISCUSS

~ 20 pages

WEEK 5

45m Geology Lessons - Lesson 85

Flex Day/Current Events

Materials: Selected geology literature, lab/field supplies, or internet access

→ NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some



Term 2

geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

WEEK 6

45m Geology Lessons - Lesson 86

How Geology Works

Materials: The Field Guide to Geology

→ READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 6

45m Geology Lessons - Lesson 87

Practical Interpretation

Materials: 101 American Geo-Sites You've Gotta See & Notebook

→ READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

WEEK 6

45m Geology Lessons - Lesson 88

Identification

Materials: rock/mineral/fossil guides

→ RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Consider taking part in a citizen science project on your field days:

- ∞ Link: Citizen Science Sand Project
- ∞ Link: Chronolog

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day. Try to plan at least 3 field trips over the course of the year: use your books or local resources for inspiration.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 6

45m Geology Lessons - Lesson 89

Geology Literature

Materials: Annals of the Former World

→ READ, NARRATE, & DISCUSS

~ 20 pages

WEEK 6

45m Geology Lessons - Lesson 90

Flex Day/Current Events

Materials: Selected geology literature, lab/field supplies, or internet access

→ NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some



Term 2

geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

WEEK 7

45m Geology Lessons - Lesson 91

How Geology Works

Materials: The Field Guide to Geology

→ READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 7

45m Geology Lessons - Lesson 92

Practical Interpretation

Materials: 101 American Geo-Sites You've Gotta See & Notebook

→ READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

WEEK 7

45m Geology Lessons - Lesson 93

Identification

Materials: rock/mineral/fossil guides

→ RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 7

45m Geology Lessons - Lesson 94

Geology Literature

Materials: Annals of the Former World

→ READ, NARRATE, & DISCUSS

~ 20 pages

WEEK 7

45m Geology Lessons - Lesson 95

Flex Day/Current Events

Materials: Selected geology literature, lab/field supplies, or internet access

→ NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some



Term 2

geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

WEEK 8

45m Geology Lessons - Lesson 96

How Geology Works

Materials: The Field Guide to Geology

→ READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 8

45m Geology Lessons - Lesson 97

Practical Interpretation

Materials: 101 American Geo-Sites You've Gotta See & Notebook

→ READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

WEEK 8

45m Geology Lessons - Lesson 98

Identification

Materials: rock/mineral/fossil guides

→ RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

→ PREP

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 8

45m Geology Lessons - Lesson 99

Geology Literature

Materials: Annals of the Former World

→ READ, NARRATE, & DISCUSS

~ 20 pages

WEEK 8

45m Geology Lessons - Lesson 100

Flex Day/Current Events

Materials: Selected geology literature, lab/field supplies, or internet access

→ NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some



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geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

WEEK 9

45m Geology Lessons - Lesson 101

How Geology Works

Materials: The Field Guide to Geology

→ READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 9

45m Geology Lessons - Lesson 102

Practical Interpretation

Materials: 101 American Geo-Sites You've Gotta See & Notebook

→ READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

WEEK 9

45m Geology Lessons - Lesson 103

Identification

Materials: rock/mineral/fossil guides

→ RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 9

45m Geology Lessons - Lesson 104

Geology Literature

Materials: Annals of the Former World

→ READ, NARRATE, & DISCUSS

~ 20 pages

WEEK 9

45m Geology Lessons - Lesson 105

Flex Day/Current Events

Materials: Selected geology literature, lab/field supplies, or internet access

→ NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some



Term 2

geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

WEEK 10 45m Geology Lessons - Lesson 106

How Geology Works

Materials: The Field Guide to Geology

→ READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

- PLAN WEEKLY
- ☐ nature walk - record observations
- ☐ science free read

WEEK 10 45m Geology Lessons - Lesson 107

Practical Interpretation

Materials: 101 American Geo-Sites You've Gotta See & Notebook

→ READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

WEEK 10 45m Geology Lessons - Lesson 108

Identification

Materials: rock/mineral/fossil guides

→ RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 10 45m Geology Lessons - Lesson 109

Geology Literature

Materials: Annals of the Former World

→ READ, NARRATE, & DISCUSS

~ 20 pages

WEEK 10 45m Geology Lessons - Lesson 110

Flex Day/Current Events

Materials: Selected geology literature, lab/field supplies, or internet access

→ NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some



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geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

WEEK 11

45m Geology Lessons - Lesson 111

How Geology Works

Materials: The Field Guide to Geology

→ READ, NARRATE, & DISCUSS

Read about 12.5 pages per week. Narrate in a way that is helpful to you. Copy any diagrams or images that are helpful.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 11

45m Geology Lessons - Lesson 112

Practical Interpretation

Materials: 101 American Geo-Sites You've Gotta See & Notebook

→ READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

WEEK 11

45m Geology Lessons - Lesson 113

Identification

Materials: rock/mineral/fossil guides

→ RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally.

WEEK 11

45m Geology Lessons - Lesson 114

Geology Literature

Materials: Annals of the Former World

→ READ, NARRATE, & DISCUSS

~ 20 pages

WEEK 11

45m Geology Lessons - Lesson 115

Present What You've Learned

→ PREPARE

For one of your exam questions next week, you will be asked to give another informal presentation to your teacher on what you have discovered during this second term about your local geology and geological history. Use this time to gather your thoughts and think through how you want to do that. You can be as creative as you would like. You will do a more formal presentation at the end of the year. This is more like an update on what you have discovered so far.

WEEK 12

45m Geology Lessons - Lesson 116

Exam Week

→ EXAMS

Answer question(s) related to course.



Term 2	
WEEK 12 45m Geology Lessons - Lesson 117 Exam Week → EXAMS Answer question(s) related to course.	
WEEK 12 45m Geology Lessons - Lesson 118 Exam Week → EXAMS Answer question(s) related to course.	
WEEK 12 45m Geology Lessons - Lesson 119 Exam Week → EXAMS Answer question(s) related to course.	
WEEK 12 45m Geology Lessons - Lesson 120 Exam Week → EXAMS Answer question(s) related to course.	



Term 3

WEEK 1

45m Geology Lessons - Lesson 121

How Geology Works

Materials: The Story of the Earth in 25 Rocks

→ READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 1

45m Geology Lessons - Lesson 122

Practical Interpretation

Materials: 101 American Geo-Sites You've Gotta See & notebook

→ READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

WEEK 1

45m Geology Lessons - Lesson 123

Identification

→ RESEARCH

Where do you plan to visit this term? Is there a specific rock type or formation that you want to learn more about? Use the resources that you discovered last term to make a plan for this term. What questions about your local geology do you have? How will you find the answers?

∞ Link: Gator Girl Rocks

∞ Link: How to Find Rocks

∞ Link: Rock Clubs by State

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day. Try to plan at least 3 field trips over the course of the year: use your books or local resources for inspiration.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 1

45m Geology Lessons - Lesson 124

How Geology Works

Materials: The Story of the Earth in 25 Rocks

→ READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

WEEK 1

45m Geology Lessons - Lesson 125

Flex Day/Current Events

Materials: Selected geology literature, lab/field supplies, or internet access

→ NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as The Map that Changed the World, or more active by pursuing a special study of interest.

OR



Term 3

→ VIEW & NARRATE

You might check out this video introduction to rocks and the geology of the Lake Superior region.

∞ Video Link: Wolf Ridge, MN

WEEK 2 45m Geology Lessons - Lesson 126

How Geology Works

Materials: The Story of the Earth in 25 Rocks

→ READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 2 45m Geology Lessons - Lesson 127

Practical Interpretation

Materials: 101 American Geo-Sites You've Gotta See & notebook

→ READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

WEEK 2 45m Geology Lessons - Lesson 128

Identification

Materials: rock/mineral/fossil guides

→ RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day. Try to plan at least 3 field trips over the course of the year: use your books or local resources for inspiration.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 2 45m Geology Lessons - Lesson 129

How Geology Works

Materials: The Story of the Earth in 25 Rocks

→ READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

WEEK 2 45m Geology Lessons - Lesson 130

Flex Day/Current Events

Materials: Selected geology literature, lab/field supplies, or internet access

→ NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some



Term 3

geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

OR

→ VIEW & NARRATE

Watch this video about the 1906 earthquake. You may remember a map in your *All Over the Map* geography book about the boats that sank in the harbor he references here. Take a look!

∞ Video Link: [The Gold Rush and the 1906 Earthquake](#)

WEEK 3 45m Geology Lessons - Lesson 131

How Geology Works

Materials: *The Story of the Earth in 25 Rocks*

→ READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 3 45m Geology Lessons - Lesson 132

Practical Interpretation

Materials: 101 American Geo-Sites You've Gotta See & notebook

→ READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

WEEK 3 45m Geology Lessons - Lesson 133

Identification

Materials: rock/mineral/fossil guides

→ RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Consider taking part in a citizen science project on your field days:

∞ Link: [Citizen Science Sand Project](#)

∞ Link: [Chronolog](#)

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 3 45m Geology Lessons - Lesson 134

How Geology Works

Materials: *The Story of the Earth in 25 Rocks*

→ READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.



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WEEK 3 45m Geology Lessons - Lesson 135

Flex Day/Current Events

Materials: Selected geology literature, lab/field supplies, or internet access

→ NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

OR

→ VIEW & NARRATE

Choose a few of these videos about our national parks.

∞ Video Link: National Parks Geology

WEEK 4 45m Geology Lessons - Lesson 136

How Geology Works

Materials: *The Story of the Earth in 25 Rocks*

→ READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

• PLAN WEEKLY

- nature walk - record observations
- science free read

WEEK 4 45m Geology Lessons - Lesson 137

Practical Interpretation

Materials: 101 American Geo-Sites You've Gotta See & notebook

→ READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

WEEK 4 45m Geology Lessons - Lesson 138

Identification

Materials: rock/mineral/fossil guides

→ RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.



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WEEK 4 45m Geology Lessons - Lesson 139

How Geology Works

Materials: The Story of the Earth in 25 Rocks

→ READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

WEEK 4 45m Geology Lessons - Lesson 140

Flex Day/Current Events

Materials: Selected geology literature, lab/field supplies, or internet access

→ NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as The Map that Changed the World, or more active by pursuing a special study of interest.

OR

→ VIEW & NARRATE

Choose a few of these videos about our national parks.

∞ Video Link: National Parks Geology

WEEK 5 45m Geology Lessons - Lesson 141

How Geology Works

Materials: The Story of the Earth in 25 Rocks

→ READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 5 45m Geology Lessons - Lesson 142

Practical Interpretation

Materials: 101 American Geo-Sites You've Gotta See & notebook

→ READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

WEEK 5 45m Geology Lessons - Lesson 143

Identification

Materials: rock/mineral/fossil guides

→ RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

→ PREP FOR FIELD DAY



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- Decide where you will explore on your field day. Try to plan at least 3 field trips over the course of the year: use your books or local resources for inspiration.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 5 45m Geology Lessons - Lesson 144

How Geology Works

Materials: The Story of the Earth in 25 Rocks

→ READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

WEEK 5 45m Geology Lessons - Lesson 145

Flex Day/Current Events

Materials: Selected geology literature, lab/field supplies, or internet access

→ NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

OR

→ VIEW & NARRATE

Spend time exploring other geology articles at this site after watching this one about Marie Tharp!

∞ Article Link: Marie Tharp's Groundbreaking Maps

WEEK 6 45m Geology Lessons - Lesson 146

How Geology Works

Materials: The Story of the Earth in 25 Rocks

→ READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

• PLAN WEEKLY

☐ nature walk - record observations

☐ science free read

WEEK 6 45m Geology Lessons - Lesson 147

Practical Interpretation

Materials: 101 American Geo-Sites You've Gotta See & notebook

→ READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

WEEK 6 45m Geology Lessons - Lesson 148

Identification

Materials: rock/mineral/fossil guides

→ RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and



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identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day. Try to plan at least 3 field trips over the course of the year: use your books or local resources for inspiration.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 6 ☐ 45m Geology Lessons - Lesson 149

How Geology Works

☐ Materials: The Story of the Earth in 25 Rocks

→ READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

WEEK 6 ☐ 45m Geology Lessons - Lesson 150

Flex Day/Current Events

☐ Materials: Selected geology literature, lab/field supplies, or internet access

→ NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as *The Map that Changed the World*, or more active by pursuing a special study of interest.

OR

→ VIEW & NARRATE

Watch this video about geology and rockfalls.

∞ Video Link: Rockfalls at Yosemite

WEEK 7 ☐ 45m Geology Lessons - Lesson 151

How Geology Works

☐ Materials: The Story of the Earth in 25 Rocks

→ READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 7 ☐ 45m Geology Lessons - Lesson 152

Practical Interpretation

☐ Materials: 101 American Geo-Sites You've Gotta See & notebook

→ READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.



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WEEK 7 45m Geology Lessons - Lesson 153

Identification

Materials: rock/mineral/fossil guides

→ RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 7 45m Geology Lessons - Lesson 154

How Geology Works

Materials: The Story of the Earth in 25 Rocks

→ READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

WEEK 7 45m Geology Lessons - Lesson 155

Flex Day/Current Events

Materials: Selected geology literature, lab/field supplies, or internet access

→ NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as The Map that Changed the World, or more active by pursuing a special study of interest.

OR

→ LISTEN/VIEW

Choose some podcast/video episodes:

∞ Audio Link: Geo Podcast

∞ Video Link: Geology Video Playlist

WEEK 8 45m Geology Lessons - Lesson 156

How Geology Works

Materials: The Story of the Earth in 25 Rocks

→ READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

- PLAN WEEKLY
 - ☐ nature walk - record observations
 - ☐ science free read



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WEEK 8 45m Geology Lessons - Lesson 157

Practical Interpretation

Materials: 101 American Geo-Sites You've Gotta See & notebook

→ READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

WEEK 8 45m Geology Lessons - Lesson 158

Identification

Materials: rock/mineral/fossil guides

→ RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 8 45m Geology Lessons - Lesson 159

How Geology Works

Materials: The Story of the Earth in 25 Rocks

→ READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

WEEK 8 45m Geology Lessons - Lesson 160

Flex Day/Current Events

Materials: Selected geology literature, lab/field supplies, or internet access

→ NOTE

Use this flex day to finish the week's reading from any of your geology books (scheduled or by your own selection), continue any lab/field activities, or read some geology current events (in your Quick Links). In this way, you may use this day to tailor the course to be more literary by reading an additional text, such as The Map that Changed the World, or more active by pursuing a special study of interest.

OR

→ VIEW & NARRATE

Watch these fascinating videos about geology, music, and art! We live in an amazing world! Art and music help us to experience and see things in new ways. The last video is a bit long, so finish it at another time if you are interested.

- ∞ Video Link: Ringing Rocks
- ∞ Video Link: Balancing Rocks 1
- ∞ Video Link: Balancing Rocks 2
- ∞ Video Link: Andy Goldsworthy



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WEEK 9 45m Geology Lessons - Lesson 161

How Geology Works

Materials: The Story of the Earth in 25 Rocks

→ READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 9 45m Geology Lessons - Lesson 162

Practical Interpretation

Materials: 101 American Geo-Sites You've Gotta See & notebook

→ READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

WEEK 9 45m Geology Lessons - Lesson 163

Identification

Materials: rock/mineral/fossil guides

→ RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 9 45m Geology Lessons - Lesson 164

How Geology Works

Materials: The Story of the Earth in 25 Rocks

→ READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

WEEK 9 45m Geology Lessons - Lesson 165

Present What You've Learned

→ PLAN

Make a plan for how you will take all the information you have researched on your local area: fieldwork, experiments, photos, illustrations, drawings, maps, and specimens, and prepare a presentation about your local geology and its history, along with any other places you visited.

You might type up notes and give your presentation as a talk; you might want to create a PowerPoint or Google Slides presentation. Or, you might choose to design something in Canva or another visual tool. You might like to create a display that uses maps and charts with your specimens laid out and labeled. You might want to design an infographic that uses your specimens to describe your local area. You decide. Be



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as creative as you would like. You can include poetry or music that you write, poetry and music that you find, and/or incorporate paintings, drawings, or illustrations.

You will have three weeks to work on it. Then, you will give your presentation to a live audience during exam week.

WEEK 10 45m Geology Lessons - Lesson 166

How Geology Works

Materials: The Story of the Earth in 25 Rocks

→ READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 10 45m Geology Lessons - Lesson 167

Practical Interpretation

Materials: 101 American Geo-Sites You've Gotta See & notebook

→ READ, NARRATE, & DISCUSS

~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.

WEEK 10 45m Geology Lessons - Lesson 168

Identification

Materials: rock/mineral/fossil guides

→ RESEARCH

Try to identify any rocks, minerals, or fossils that you have picked up locally. Ask local experts to see their collections. Ask them for help and direction in finding and identifying what is readily available locally. Ask for recommendations for identification guides that are appropriate for your area and that they have found helpful.

→ PREP FOR FIELD DAY

- Decide where you will explore on your field day.
- Check the weather and confirm when you will go.
- Collect any tools or materials that you might want to take with you.

WEEK 10 45m Geology Lessons - Lesson 169

How Geology Works

Materials: The Story of the Earth in 25 Rocks

→ READ, NARRATE, & DISCUSS

2-3 chapters per week; note that you have 2 lesson days for this book.

WEEK 10 45m Geology Lessons - Lesson 170

Present What You've Learned

→ PREPARE

Continue to prepare your presentation.



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WEEK 11 45m Geology Lessons - Lesson 171 <i>How Geology Works</i> Materials: The Story of the Earth in 25 Rocks → READ, NARRATE, & DISCUSS 2-3 chapters per week; note that you have 2 lesson days for this book.	• PLAN WEEKLY nature walk - record observations science free read
WEEK 11 45m Geology Lessons - Lesson 172 <i>Practical Interpretation</i> Materials: 101 American Geo-Sites You've Gotta See & notebook → READ, NARRATE, & DISCUSS ~ 2-5 sites per week, depending on how much time you want to spend with each one. You might research the type(s) of rocks found at the site and/or examine the geological maps of the site. As you read, sketch a map of the US that shows the location of the sites and what is there.	
WEEK 11 45m Geology Lessons - Lesson 173 <i>Identification</i> Materials: rock/mineral/fossil guides → RESEARCH Try to identify any rocks, minerals, or fossils that you have picked up locally.	
WEEK 11 45m Geology Lessons - Lesson 174 <i>How Geology Works</i> Materials: The Story of the Earth in 25 Rocks → READ, NARRATE, & DISCUSS 2-3 chapters per week; note that you have 2 lesson days for this book.	
WEEK 11 45m Geology Lessons - Lesson 175 <i>Present What You've Learned</i> → PREPARE Continue to prepare your presentation. Enjoy giving it next week to a live audience!	
WEEK 12 45m Geology Lessons - Lesson 176 <i>Exam Week</i> → EXAMS Answer question(s) related to course.	
WEEK 12 45m Geology Lessons - Lesson 177 <i>Exam Week</i> → EXAMS Answer question(s) related to course.	



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WEEK 12 45m Geology Lessons - Lesson 178 Exam Week → EXAMS Answer question(s) related to course.	
WEEK 12 45m Geology Lessons - Lesson 179 Exam Week → EXAMS Answer question(s) related to course.	
WEEK 12 45m Geology Lessons - Lesson 180 Exam Week → EXAMS Answer question(s) related to course.	

Geology Lessons

Examination

Term 1

- Field Guide to Geology: Explain in what ways the earth's crust changes.
- Field Guide to Geology: Describe different ways that rocks are formed through the changes in the earth's crust.
- Explain how you identified a found rock this term. OR Tell about a geology current event.
- Annals of a Former World: Explain some aspect of geology that McPhee has brought to life for you.
- Explain what you have discovered so far about your local geology and geological history. Share your notebook.

Term 2

- Field Guide to Geology: Describe how water shapes the land through rivers OR seas OR ice.
- Field Guide to Geology: Explain how relative dating works.
- Explain how you identified a found rock this term. OR Tell about a geology current event.
- Annals of a Former World: Explain some aspect of geology that McPhee has brought to life for you.
- Tell what you have discovered this term about your local geology and geological history. Share your notebook.

Term 3

- Tell the story of one of the rocks from The Story of the Earth in 25 Rocks.
- The Story of the Earth in 25 Rocks: Explain how a geologist is like a detective.
- Explain how you identified a found rock this term. OR Tell about a geology current event.
- Tell about a geological site that you read about that you would like to visit someday. Where is it? What is interesting about it? What would you hope to see or learn?
- Give a presentation on your local geology and geological history. Share your notebook.